DDADA
AEBAA
BCEBB
BDDCA

1.4 comparative advantage

1.

	Brazil	Spain
Corn	2 hours of labor	4 hours of labor
Shoes	8 hours of labor	12 hours of labor

The table above shows the amount of labor required to produce a unit of corn and a unit of shoes in Brazil and Spain. If both countries have equal numbers of workers, what pattern of international trade between Brazil and Spain is most likely to emerge?

(A) Brazil will export both corn and shoes.

(B) Spain will export both corn and shoes.

(C) No mutually beneficial trade can occur.

(D) Brazil will export corn and import shoes. ✓

(E) Brazil will import corn and export shoes.

2. Dana and Robin produce smoothies and pizza. In one hour Dana can make 20 smoothies or 10 pizzas. In one hour Robin can make 18 smoothies or 6 pizzas. Which of the following statements is true?

(A) Robin has an absolute advantage in making smoothies and a comparative advantage in making pizzas.

(B) Robin has both an absolute advantage and a comparative advantage in making pizzas.

(C) Dana has a comparative advantage in making both smoothies and pizzas.

(D) Dana has a comparative advantage in making pizzas, and Robin has a comparative advantage in making smoothies. ✓

(E) Dana has a comparative advantage in making smoothies, and Robin has a comparative advantage in making pizzas.

3. Nation Alpha can produce either 3 units of good X or 1 unit of good Y with one hour of labor, whereas nation Beta can produce either 4 units of good X or 2 units of good Y with one hour of labor. Assuming that labor is the only input, which of the following is true?

(A) Beta has an absolute advantage in the production of good X. ✓

(B) Alpha has an absolute advantage in the production of good Y.

(C) Beta has a comparative advantage in the production of good X.

(D) Alpha has a comparative advantage in the production of good Y.

(E) Alpha has both an absolute and comparative advantage in the production of good Y.

Answer A

Correct. Beta has an absolute advantage in the production of good X given that it can produce more X in

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an hour than Alpha can (4 is greater than 3).

4. Cindy and Martin both sew t-shirts in a small factory. Using the same resources, Martin can sew twelve t-shirts and Cindy can sew nine t-shirts in one day. Which of the following can be concluded from the given information?

(A) Martin has a higher opportunity cost of sewing t-shirts than Cindy does.

(B) Cindy has a comparative advantage in sewing t-shirts.

(C) Martin has a comparative advantage in sewing t-shirts.

(D) Martin has an absolute advantage in sewing t-shirts. ✓

(E) Cindy has an absolute advantage in sewing t-shirts.

Answer D

Correct. Martin has an absolute advantage in sewing t-shirts because he can sew more t-shirts than Cindy $(12 > 9)$ using the same resources.

5. A country can consume beyond its present production possibilities curve when it

(A) trades with other countries, thus taking advantages of different opportunity costs ✓

(B) reduces unemployment, thus increasing output

(C) experiences decreasing opportunity costs.

(D) faces an upward-sloping production possibilities curve

(E) faces a downward-sloping production possibilities curve

6. A farmer in Country A can harvest 20 bushels of wheat or 10 bushels of corn in a day, while a farmer in Country B can harvest 8 bushels of wheat or 8 bushels of corn in a day. If Country A and Country B specialize and trade, Country A will

(A) export wheat and import corn ✓

(B) export corn and import wheat

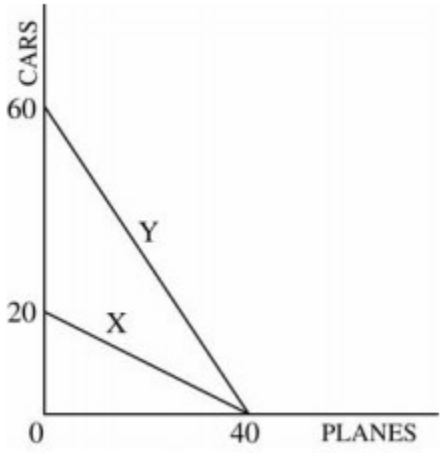
(C) export both wheat and corn

(D) import both wheat and corn

(E) benefit more from trade than will Country B

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7.



The diagram above shows the production possibilities curves for two countries, Country X and Country Y. Assume that both countries use equal amounts of resources in production. If the two countries engage in trade, both would be better off under which of the following conditions?

(A) Country X produced both cars and planes, because it has an absolute advantage in the production of both goods.

(B) Country Y produced both cars and planes, because it has a comparative advantage in the production of both goods.

(C) Country X specialized in the production of cars, because it has an absolute advantage in the production of cars.

(D) Country X specialized in the production of cars, because it has a comparative advantage in the production of cars.

(E) Country Y specialized in the production of cars, because it has a comparative advantage in the production of cars. ✓

Both Amy and Brad produce and consume apple pie and ice cream. In one hour, Brad makes five apple pies or ten gallons of ice cream while Amy makes fifteen apple pies or fifteen gallons of ice cream.

8. Based on the information, one can correctly conclude that

(A) Brad has an absolute advantage in making ice cream and will sell ice cream to Amy

(B) Brad has a comparative advantage in making ice cream and will sell ice cream to Amy ✓

(C) Brad does not have a comparative advantage in making either good and will not trade with Amy

(D) Brad has an absolute advantage in making apple pies but a comparative advantage in making ice cream

(E) Brad's opportunity cost of making ice cream is higher than Amy's

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9. A professor hires two aides, assigning them the tasks of reading student papers and of typing lecture notes on a computer. One of the aides, Ben, can read 1 page of a student paper per minute or type 50 words of lecture notes per minute, and the other aide, Ann, can read 3 pages of a student paper per minute or type 60 words of lecture notes per minute.

Which of the following statements is true?

(A) Ann has a comparative advantage in reading student papers, and Ben has a comparative advantage in typing lecture notes. ✓

(B) Ann has an absolute advantage in reading student papers, and Ben has an absolute advantage in typing lecture notes.

(C) Ben has a comparative advantage in reading student papers, and Ann has a comparative advantage in typing lecture notes.

(D) Ben has a comparative advantage in both reading student papers and typing lecture notes.

(E) Ben has an absolute advantage in both reading student papers and typing lecture notes.

10. Assume both Italy and Greece produce only two goods: wine and olive oil. If Italy holds a comparative advantage in the production of wine, then which of the following statements is NOT true?

(A) Italy must hold an absolute advantage in the production of wine. ✓

(B) Greece holds a comparative advantage in the production of olive oil.

(C) Italy's opportunity cost of producing one additional unit of wine is lower than Greece's.

(D) Greece's opportunity cost of producing one additional unit of olive oil is lower than Italy's.

(E) If trade is open between them, these countries have an incentive to trade.

11.

UNITS OF MANUFACTURED GOODS AND SERVICE GOODS PRODUCED USING ONE HOUR OF LABOR

Country	Manufactured Goods	Service Goods
A	100 units	300 units
B	75 units	150 units

According to the information in the table above, which of the following statements is true if both countries have the same number of workers?

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(A) Country A has both an absolute and a comparative advantage in manufactured goods.

(B) Country A has an absolute advantage in manufactured goods but a comparative advantage in service goods. ✓

(C) Country B has a comparative advantage in service goods but no absolute advantage in either good.

(D) Country A has an absolute advantage in service goods but a comparative advantage in manufactured goods.

(E) Country B has an absolute advantage in manufactured goods, but without more information, it is not possible to tell in which product it has a comparative advantage.

12. The table provided shows the number of labor hours required to produce a cell phone or a unit of lumber in Estonia and in Finland.

	Cell Phone	Unit of Lumber
Estonia	20 hours	5 hours
Finland	20 hours	4 hours

Based on the information in the table, which of the following is true?

(A) Estonia has a comparative advantage in producing lumber.

(B) Finland has a comparative advantage in producing cell phones.

(C) Finland has an absolute advantage in producing lumber. ✓

(D) Estonia has both the absolute and comparative advantages in producing lumber.

(E) Neither country has a comparative advantage in producing cell phones.

Answer C

Correct. Finland can produce a unit of lumber with fewer hours of labor than Estonia and therefore has an absolute advantage in producing lumber.

Country	Opportunity Cost of 1 Ton of Apples	Opportunity Cost of 1 Ton of Oranges
X	1 ton of oranges	1 ton of apples
Y	2 tons of oranges	0.5 ton of apples

The table above shows the opportunity costs of producing apples and oranges in Countries X and Y.

13. Which of the following can be concluded based on the data given in the table?

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(A) Country Y has an absolute advantage in producing both goods.
(B) Country Y has a comparative advantage in producing both goods.
(C) Country X has an absolute advantage in producing both goods.
(D) Country X has a comparative advantage in producing oranges.
(E) Country X has a comparative advantage in producing apples. ✓

Answer E

Correct. Comparative advantage is determined by differences in opportunity costs in performing a particular task or producing a product or service. The opportunity cost of producing 1 ton of apples is 1 ton of oranges in Country X and 2 tons of oranges in Country Y. Country X has a lower opportunity cost than Country Y in producing apples. Therefore, Country X has a comparative advantage in producing apples.

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14. Dr. Art and Dr. Zed jointly own an outpatient surgery practice. Dr. Art takes 2 hours to perform knee surgery and 1 hour to perform elbow surgery. Dr. Zed takes 4 hours to perform knee surgery and 1 hour to perform elbow surgery. Which of the following conclusions is supported by the information provided?

(A) Dr. Art has a comparative advantage in elbow surgery.
(B) Dr. Zed has a comparative advantage in elbow surgery. ✓
(C) Dr. Zed does not have a comparative advantage in performing either procedure.
(D) Dr. Art does not have a comparative advantage in performing either procedure.
(E) Dr. Zed has an absolute advantage in performing both procedures.

Answer B

Correct. The individual with the lower opportunity costs in performing elbow surgery has the comparative advantage in the provision of that service. In this case, Dr. Art can either provide 4 knee surgeries or 8 elbow surgeries in an eight-hour day (for an opportunity cost of performing an additional elbow surgery is $4/8 = 1/2$ knee surgery). Dr. Zed can either provide 2 knee surgeries or 8 elbow surgeries in an eight-hour day (for an opportunity cost of performing an additional elbow surgery is $2/8 = 1/4$ knee surgery).

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15. The chart below shows the number of resource units the countries Gamma and Omega must use to produce one unit of food or one unit of clothing.

	Gamma	Omega
Food	10 resource units	5 resource units
Clothing	9 resource units	3 resource units

Which of the following statements is true according to the chart above?

(A) Trade should take place, with Gamma specializing in clothing production and Omega specializing in food production.

(B) Trade should take place, with Gamma specializing in food production and Omega specializing in clothing production. ✓

(C) Each country should devote half of its resources to food production and half to clothing production and not trade.

(D) Since Omega can produce both food and clothing using fewer resources than Gamma, it cannot benefit from trade with Gamma

(E) Since Gamma can produce both food and clothing using fewer resources than Omega, it cannot benefit trade with Omega.

16. If nations specialize according to their comparative advantage and engage in international trade with each other, each nation can

(A) produce outside its production possibilities curve

(B) consume outside its production possibilities curve ✓

(C) shift its production possibilities curve to the right

(D) become more self-sufficient

(E) produce more of all goods

17. Which of the following statements regarding specialization and trade is true?

(A) With specialization and trade, a country's production possibilities curve shifts out, showing an increase in productive capacity.

(B) With specialization and trade, a country will produce a combination of goods beyond its production possibilities curve.

(C) To gain from trade, countries specialize in producing goods and services in which they have an absolute advantage.

(D) With specialization and trade, a country can consume beyond its production possibilities curve. ✓

(E) A country must have both the comparative and the absolute advantage in a product to specialize and gain from trade.

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Answer D

Correct. With specialization and trade a country remains on its production possibilities curve but produces more of the good for which it has comparative advantage. Trading some of that good for another good would allow a country to consume beyond its production possibilities curve.

18. Using the same amount of time and resources, Jack can assemble either 10 bikes or 5 computers, whereas Sam can assemble either 5 bikes or 5 computers. Based on the data, which of the following statements is correct?

(A) Sam has an absolute advantage in assembling bikes.
(B) Sam has an absolute advantage in assembling computers.
(C) Sam has a comparative advantage in assembling bikes.
(D) Jack has a comparative advantage in assembling bikes. ✓
(E) Jack has a comparative advantage in assembling both bikes and computers.

19. The table below shows the number of hours it takes for Kim to install a computer or to paint a garage and for Maria to install a computer or to paint a garage.

	Kim	Maria
Install computer	2 hours	4 hours
Paint garage	10 hours	12 hours

If tasks are assigned according to comparative advantage, which of the following is true?

(A) Kim should perform both tasks.
(B) Maria should perform both tasks.
(C) Kim should install computers and Maria should paint garages. ✓
(D) Kim should paint garages and Maria should install computers.
(E) Kim and Maria should both install computers.

	Labor Hours Required to Produce	
	One Pair of Shoes	One Loaf of Bread
Mr. X	5 hours	1 hour
Ms. Y	2 hours	0.5 hour

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20. According to the data, if Mr. X and Ms. Y have the opportunity to trade, which of the following will be mutually beneficial?

(A) Ms. Y sells shoes to and buys bread from Mr. X. ✓

(B) Ms. Y sells bread to and buys shoes from Mr. X.

(C) Ms. Y sells both bread and shoes to Mr. X.

(D) Ms. Y buys both bread and shoes from Mr. X.

(E) Ms. Y and Mr. X do not trade with each other.

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21. If Nation X produces coffee at a higher opportunity cost than Nation Y, which of the following is true?

(A) Nation X must have an absolute advantage in producing coffee.

(B) Nation X must have a comparative advantage in producing coffee.

(C) Nation Y must have an absolute advantage in producing coffee.

(D) Nation Y must have a comparative advantage in producing coffee. ✓

(E) There is insufficient information to determine both absolute and comparative advantage.

Answer D

Correct. Given that Nation Y has a lower opportunity cost than Nation X , it must have a comparative advantage.

The table below shows the total labor hours required to produce a unit of gold and the total labor hours required to produce a unit of sugar in Portugal and Spain, respectively.

	Portugal	Spain
Gold	1 hour	2 hours
Sugar	4 hours	3 hours

22. Which country has an absolute advantage in sugar production?

(A) Neither Spain nor Portugal, because both require fewer labor hours to produce a unit of gold than to produce a unit of sugar.

(B) Spain, because the opportunity cost of sugar in terms of gold is lower.

(C) Spain, because it requires fewer total labor hours than Portugal to produce a unit of sugar. ✓

(D) Portugal, because it requires more total labor hours than Spain to produce a unit of sugar.

(E) Portugal, because it takes only a quarter of an hour to produce a unit of sugar.

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23.

	Palm Leaves	Coconuts
Robert	8	2
Frank	10	5

The table above shows the maximum number of palm leaves or coconuts that Robert and Frank can pick respectively in a single day. Which of the following is true?

(A) Robert has a comparative advantage in picking coconuts.

(B) Frank has a comparative advantage in picking palm leaves.

(C) Robert and Frank can both benefit from trade with each other if 1 coconut is traded for 1 palm leaf.

(D) Robert and Frank can both benefit from trade with each other if 1 coconut is traded for 3 palm leaves. ✓

(E) Robert and Frank can both benefit from trade with each other if 1 coconut is traded for 5 palm leaves.

24. Suppose that in one week Sam can knit 5 sweaters or make 4 blankets and Rob can knit 10 sweaters or make 6 blankets. Which of the following is true?

(A) Sam has an absolute advantage in making blankets.

(B) Sam has an absolute advantage in knitting sweaters.

(C) Sam has a comparative advantage in making blankets. ✓

(D) Sam has a comparative advantage in knitting sweaters.

(E) Sam has neither a comparative nor an absolute advantage in knitting sweaters or making blankets.

25. Nation Aga can produce either 3 units of good X or 1 unit of good Y with one hour of labor, while nation Kaza can produce either 4 units of good X or 2 units of good Y with one hour of labor. Assuming that labor is the only input, mutually beneficial exchange can take place between Aga and Kaza if

(A) Aga exchanges $\frac{1}{2}$ unit of good X for 1 unit of good Y from Kaza.

(B) Aga exchanges $1\frac{1}{2}$ units of good X for 1 unit of good Y from Kaza.

(C) Aga exchanges $2\frac{1}{2}$ units of good X for 1 unit of good Y from Kaza. ✓

(D) Kaza exchanges 1 unit of good Y for 1 unit of good X from Aga.

(E) Kaza exchanges 2 units of good Y for 1 unit of good X from Aga.

Answer C

Correct. Kaza's opportunity cost of producing 1 unit of Y is 2 units of X and Aga's opportunity cost of producing 1 unit of Y is 3 units of X; thus, exchanging 1 unit of Y for $2\frac{1}{2}$ units of X is beneficial to both nations.

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26.

	Orange juice (in liters)	Canvases
Stefan	40	30
Donna	20	10

Stefan and Donna produce orange juice and canvases. The table provided shows the maximum possible production of orange juice OR canvases that can be produced by Stefan and Donna. Which of the following terms of trade would be mutually beneficial to Stefan and Donna?

(A) 1 canvas for 0.4 liter of orange juice

(B) 1 canvas for 0.5 liter of orange juice

(C) 1 canvas for 1 liter of orange juice

(D) 1 canvas for 1.5 liters of orange juice ✓

(E) 1 canvas for 2.5 liters of orange juice

Answer D

Correct. Trade is mutually beneficial when the terms of trade fall between the opportunity costs for the two people (or nations) involved. In this case, Stefan's opportunity cost of producing one liter of orange juice is $\frac{4}{3} = 1.33$ canvases and Donna's opportunity cost of producing one liter of orange juice is $\frac{20}{10} = 2$ canvases. Since a term of trade of 1 canvas for 1.5 liters of orange juice falls between the opportunity costs for Stefan and Donna, then trade would be mutually beneficial.

27. In the country Beta it takes two hours of labor to produce a unit of tea and four hours of labor to produce a unit of bread. In the country Zeta it takes three hours of labor to produce a unit of tea and nine hours of labor to produce a unit of bread. Which of the following is true?

(A) Country Zeta has an absolute advantage in producing both tea and bread.

(B) Country Zeta has an absolute advantage and a comparative advantage in producing tea.

(C) Country Beta has an absolute advantage and a comparative advantage in producing tea.

(D) Country Beta has the lower opportunity cost in producing bread and therefore a comparative advantage in producing bread. ✓

(E) Country Beta, with trade, will export both tea and bread to country Zeta.

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Answer D

Correct. A country has a comparative advantage in producing a good if it has the lowest opportunity cost of producing the good. In Beta it takes 2 hours to produce a unit of tea and 4 hours to produce a unit of bread; the opportunity cost of a unit of bread is two units of tea. In Zeta it takes 3 hours to produce a unit of tea and 9 hours to produce a unit of bread; the opportunity cost of a unit of bread is 3 units of tea. Since two teas (Beta) is less than three teas (Zeta) for a unit of bread, Beta has the comparative advantage in producing bread.